5(a)	A photon is a quantum of electromagnetic radiation with energy given by *hf*, where *h* is the planck's constant and *f* is the frequency of the wave.
5(b)(i)	$E = hf = \dfrac{hc}{\lambda}$ $\Delta E = E_f - E_i$ $= \left(6.63\times10^{-34}\right)\left(3.00\times10^{8}\right)\left(\dfrac{1}{966.8\times10^{-12}} - \dfrac{1}{965.0\times10^{-12}}\right)$ $= -3.84\times10^{-19}\ \text{J}$
(ii)	loss in energy of photon = gain in kinetic energy of electron $3.84\times10^{-34} = \dfrac{p^2}{2\left(9.11\times10^{-31}\right)}$ (kinetic energy of electron = $\dfrac{p^2}{2m_e}$) $p = 8.36\times10^{-25}\ \text{N s}$
5(c)	Initial momentum of photon is in the horizontal direction. By conservation of momentum, the vertical component of the net final momentum of the deflected photon and electron should be zero. momentum of a photon = $\dfrac{h}{\lambda}$ vertical component = $\left(\dfrac{6.63\times10^{-34}}{966.8\times10^{-12}}\right)\sin 75^\circ = 6.62\times10^{-25}\ \text{N s}$ (upwards)

	vertical component of the momentum of deflected electron = $\left(8.36\times10^{-25}\right)\sin\alpha$ (downwards) Since net vertical component of momentum must be zero, $6.62 \times 10^{-25} = 8.36 \times 10^{-25} \sin \alpha$ $\alpha = 52.4°$
6 (a) (i)	Acceleration is the rate of change of velocity.